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Computers:

TOTAL GRAPHICS POWER... Is The MOST SIGNIFICANT ELEMENT When Designing A PCB

Computers are taken so much for granted that even engineers occasionally fail to appreciate the factors that affect their product design. This interview should give you some insight into how motherboards and graphics cards are designed. Jackson Hsu, product management director, GIGABYTE Technology Co. Ltd speaks with Dilin Anand of EFY

Q. What design elements are used to improve performance in an over-clocked enthusiast motherboard?

A. Engineers have been using glass-fabric printed circuit board (PCB) to help protect against electrical shorts due to humidity. It works by reducing the gaps between PCB filaments. They also use 2x copper PCBs in their designs, which provide sufficient power trace paths between components to handle greater than normal power loads and to remove heat from the critical central processing unit (CPU) power delivery area. This is essential to ensure the motherboard is able to handle the increased power loading that is necessary when over-clocking.

All-digital power delivery is another improvement where digital controllers offer better precision in delivering power to the mother-

board's most power-hungry and energy-sensitive components, allowing users to get the absolute maximum performance from their CPU.

You may also see additional CPU fan pin headers located in various locations on the motherboard to connect a water pump and configured for continuous full-speed operation.

Q. What design improvements have been made to improve motherboard reliability?

A. In a technology called DualBIOS, the motherboard has both a main BIOS and a backup BIOS, making you protected from BIOS failure due to hardware malfunction, improper over-clocking settings or power failure during the update process.

Motherboards integrate the best-quality solid-state capacitors

(we use long life-span, durable, black, solid capacitors) that are rated to perform at maximum efficiency for a guaranteed minimum of 10,000 hours and provide ultra-low erythrocyte sedimentation rate (ESR), no matter how high the CPU loading. These come in customised jet black from Nippon Chemi-Con and Nichicon.

6x (30µ) gold-plated CPU sockets are also used to improve the longevity of the CPU socket over time, without having any concerns about corroded pins and bad contacts. Plating thickness here is 15 times more than the industry standard.

Q. How do you improve electro-static discharge (ESD) protection in modern motherboards?

A. High-resistance integrated circuits (ICs) help protect the